Project Report

Project Name: Semester Project Management

Name	Registration number	Semester	Section	Assigned Functionalities	
Abdullah (Group Leader)	Sp23-Bse-116	5	A	Group	
Rana Asad Ur Rahman	Sp23-Bse-029	5	A	Project	
Syed Mustafa	Sp23-Bse-015	5	A	Student	
Zarak Khan	Sp23-Bse-031	5	A	Admin	
Abdul Haseeb	Sp23-Bse-002	5	A	Document	
Aizaz Afridi	Sp23-Bse-003	5	A	Login	
Ahmed Ayyar Khan	Sp23-Bse-021	5	A	Supervisor	

Abstract

The Fina Year Project (FYP) Management System" project aims to streamline the process of managing final year projects in educational institutions. The project's primary objective is to develop a comprehensive software solution that facilitates the efficient coordination, monitoring, and evaluation of final year projects.

The need for this project arises from the challenges faced in traditional manual management systems, including difficulties in project allocation, group management, and assessment. By leveraging technology, our goal is to address these challenges and enhance the overall management experience for students, faculty members, and administrators involved in the FYP process.

The project follows a systematic approach, beginning with requirement analysis and culminating in the development of a user-friendly web-based platform. Key features of the system include project proposal submission, supervisor/advisor allocation, group management, report generation, and evaluation management.

Through the implementation of this system, we anticipate several benefits, including improved transparency, enhanced communication, streamlined workflows, and better decision-making. Additionally, the system's reporting capabilities will provide valuable insights into project progress and performance.

In conclusion, the FYP Management System represents a significant advancement in project management practices within educational institutions. By embracing technology and innovation, we aim to optimize the FYP process, ultimately leading to better outcomes for students and faculty members involved in final year projects.

Introduction to the Problem

1.1 Introduction

Managing final year projects is a critical yet complex task for educational institutions. Current practices often involve manual processes or outdated systems, leading to inefficiencies, miscommunication, and difficulty in tracking progress. The need for a modern, centralized system is paramount to ensure smooth coordination and management.

1.2 Purpose

The purpose of this project is to develop an FYP Management System that provides a centralized platform for students, supervisors, and administrators. The system aims to:

- Automate project submission and evaluation processes.
- Enhance collaboration between stakeholders.
- Improve monitoring and reporting of project progress.

1.3 Objectives

The objectives of the project are:

- To design a user-friendly platform for FYP management.
- To enable features such as student and advisor registration, project submission, and evaluation tracking.
- To improve transparency, decision-making, and communication.
- To offer comprehensive reporting and analytics for better insights.

1.4 Existing Solution

Current systems rely on manual methods like paper-based records or fragmented digital tools. These methods result in:

- Difficulty in project tracking.
- Inefficient communication between students and supervisors.
- Time-consuming evaluation and reporting processes.

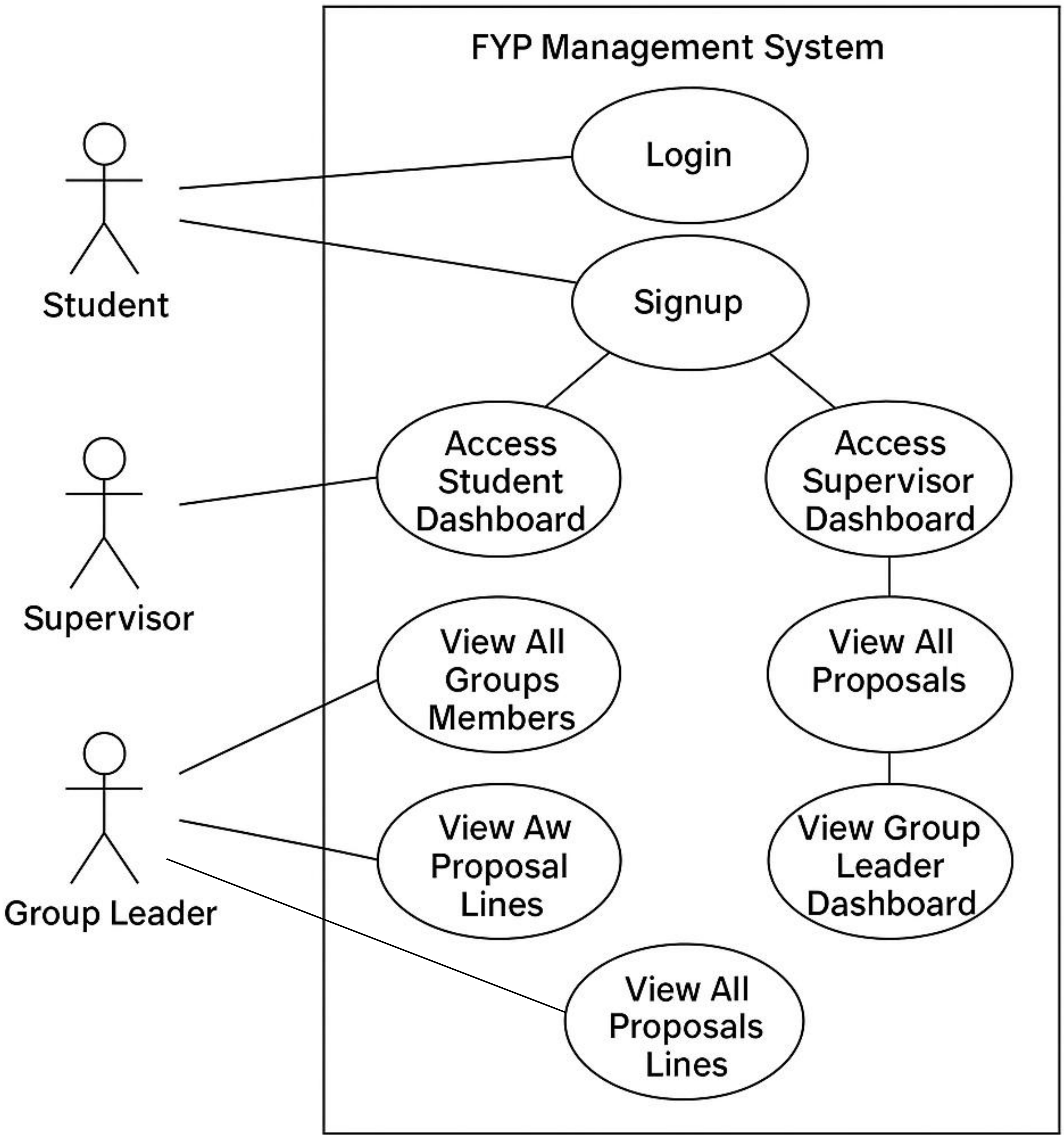
1.5 Proposed Solution

The proposed FYP Management System addresses these issues by providing a centralized, web- based solution. Features include:

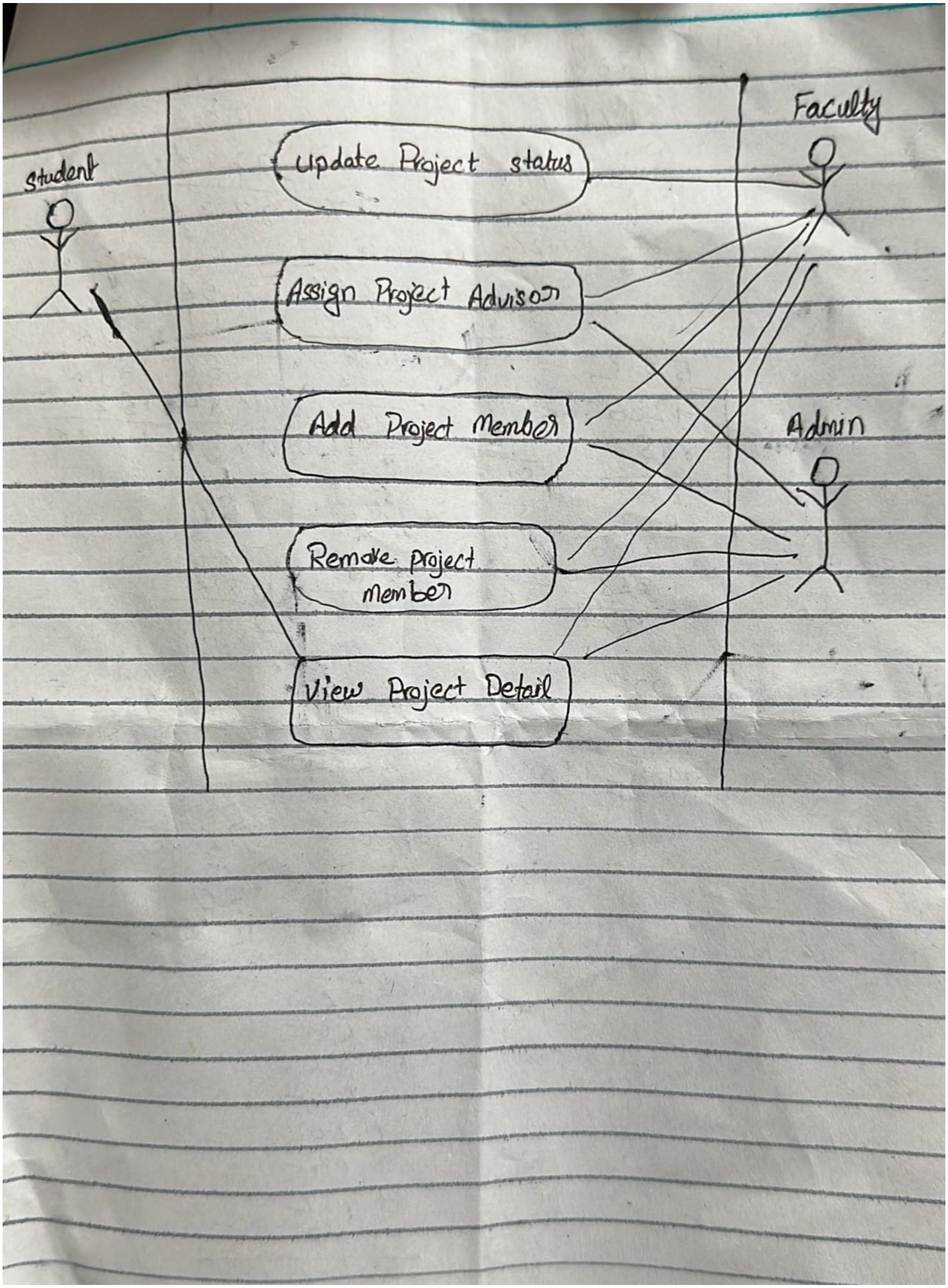
- Student and supervisor registration.
- Online project proposal submission.
- Group formation and supervisor assignment.
- Evaluation and marks management.
- Real-time progress tracking and reporting.

Chapter 1

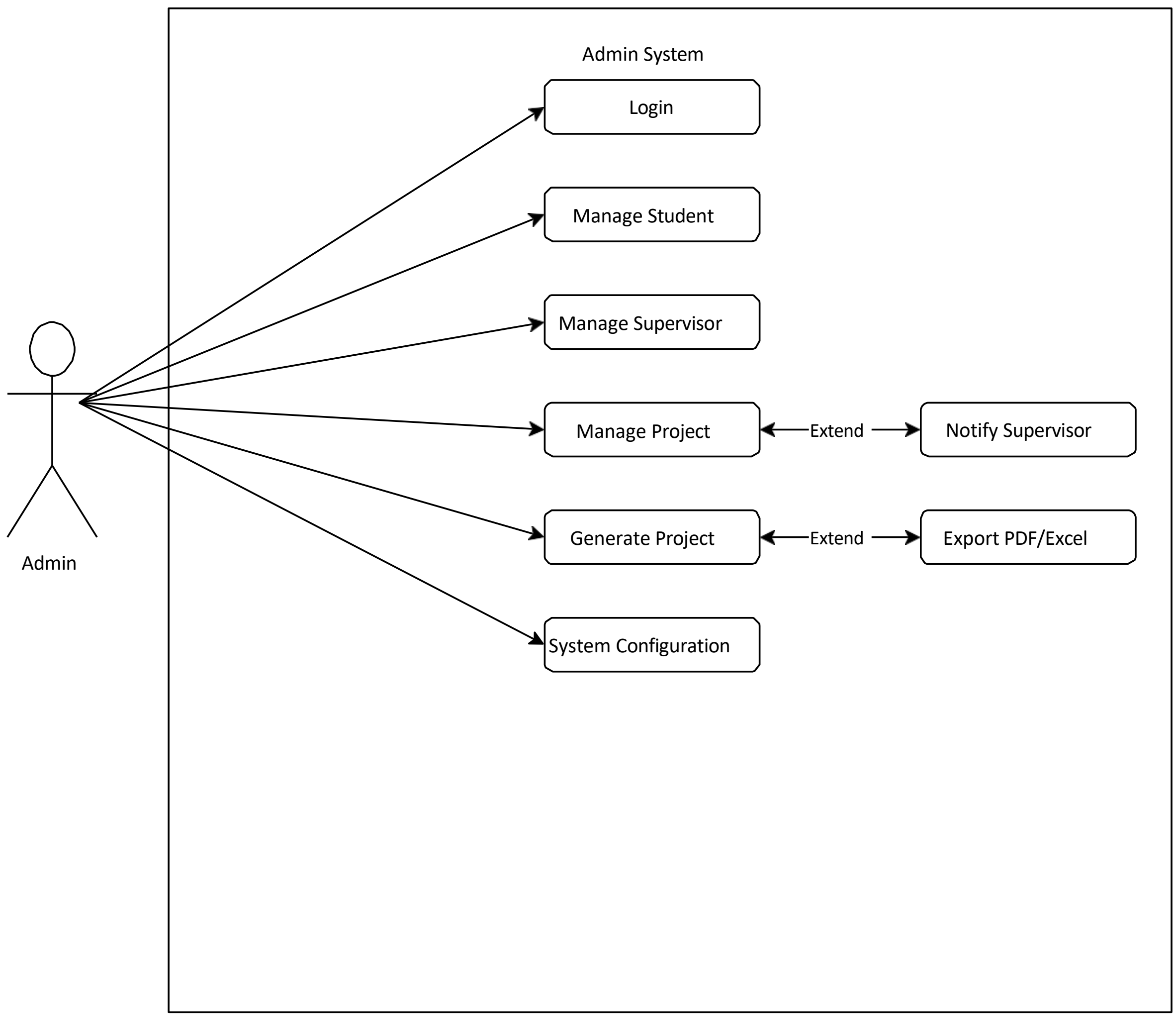
Use Case by mustafa



Rana Asad

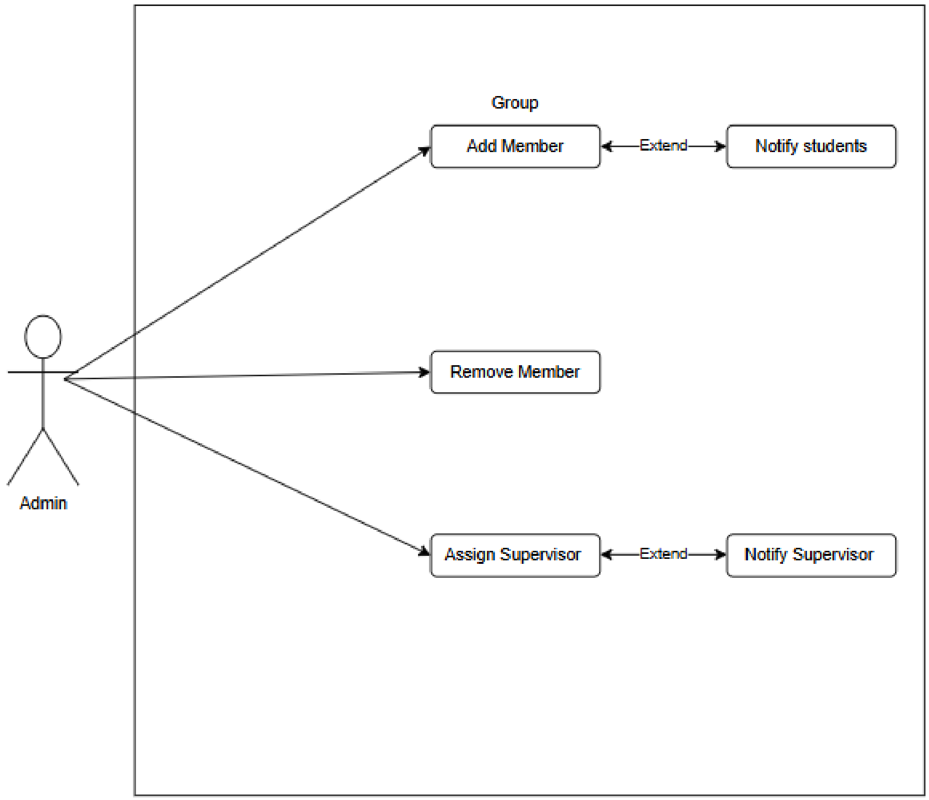

Faculty
Student
Update Project status
Assign Project Advisor
Add Project Member
Admin
Remove project member
View Project Detail

Zarak Khan



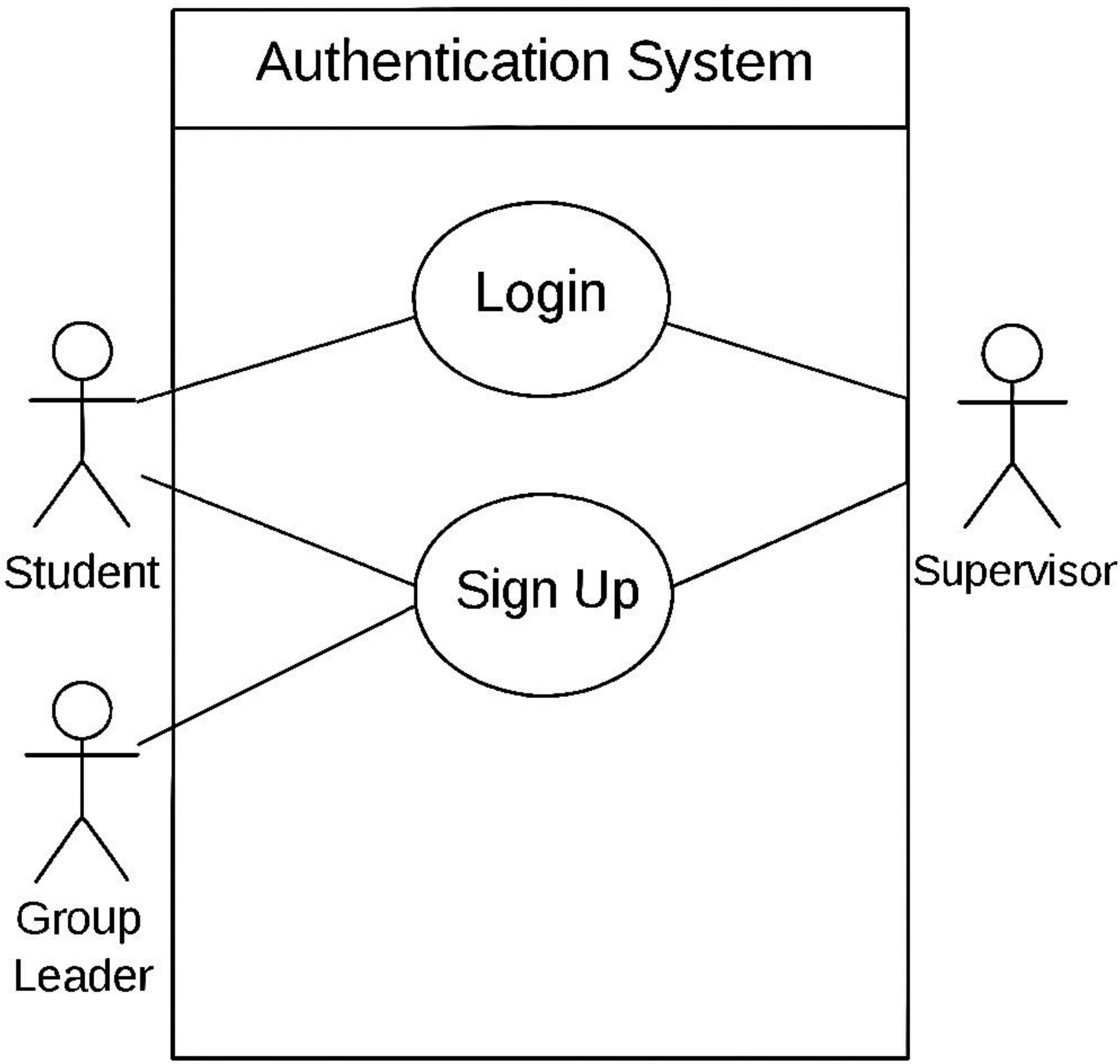
Admin System
Login
Manage Student
Manage Supervisor
Manage Project
Extend
Notify Supervisor
Generate Project
Extend
Export PDF/Excel
System Configuration
Admin

Abdullah

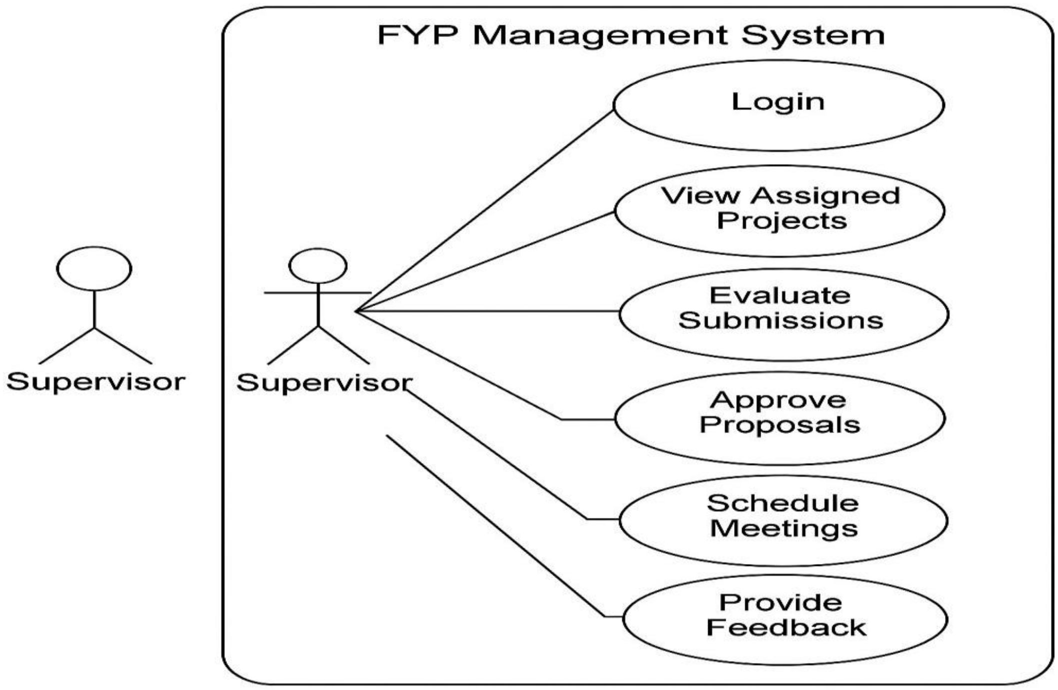


Group
Add Member
Extend
Notify students
Remove Member
Admin
Assign Supervisor
Extend
Notify Supervisor

Aizaz ULLAH USE-CASE Diagram



Ahmad Ayyar usecase



CHAPTER:2

1. Login Controller Use Cases

Use Case 1: User Login

Element	**Description**
Name	User Login
Scope	Authentication System
Level	User Goal
Primary Actor	Student, Group Leader, Supervisor
Stakeholders	Users (need secure access), System (needs to authenticate)
Preconditions	1. User exists in users.txt 2. Login view is accessible
Success Postconditions	1. User authenticated 2. Appropriate dashboard opened
Failure Postconditions	1. Login view remains open 2. Error message displayed
Main Success Scenario	1. User opens LoginView 2. Selects role, enters ID/password 3. Clicks login button 4. AuthService validates credentials 5. System opens role-specific dashboard
Extensions	3a. Invalid credentials - show error 3b. Missing fields - prompt completion 3c. File error - log and show system error
Special Requirements	1. Case-sensitive matching 2. users.txt must exist

Use Case 2: User Signup

Element	**Description**
Name	User Signup
Scope	Authentication System
Level	User Goal
Primary Actor	New User
Stakeholders	System (needs valid registrations)
Preconditions	1. User doesn't exist 2. Signup view accessible
Success Postconditions	1. New user created 2. Success message shown
Main Success Scenario	1. User enters details 2. AuthService creates new entry 3. Success confirmation shown
Extensions	2a. Duplicate ID - show error 2b. Invalid format - show requirements

2. Group Controller Use Cases

Use Case 3: View All Groups

Element	**Description**
Name	View All Groups
Scope	Group Management
Level	User Goal
Primary Actor	Supervisor, Group Leader
Preconditions	1. User authenticated 2. group.txt exists
Success Postconditions	Group list displayed
Main Success Scenario	1. User requests groups 2. System reads group.txt 3. Returns all group entries
Extensions	2a. File missing - show empty state 2b. Corrupt data - skip bad entries

Use Case 4: View Group Members

Element	**Description**
Name	View Group Members
Scope	Group Management
Level	Subfunction
Primary Actor	Supervisor, Group Leader
Preconditions	1. Valid group ID provided
Main Success Scenario	1. Finds group by ID 2. Parses comma-separated members 3. Returns member list
Extensions	1a. Group not found - return empty

3. Proposal Controller Use Cases

Use Case 5: View All Proposals

Element	Description
Name	View All Proposals
Scope	Proposal Management
Level	User Goal
Primary Actor	Supervisor, Group Leader
Preconditions	1. proposal.txt exists
Main Success Scenario	1. Reads proposal.txt 2. Parses each line to Proposal objects 3. Returns list
Extensions	1a. File missing - empty list 2a. Parse fails - skip entry

Use Case 6: View Raw Proposals

Element	Description
Name	View Raw Proposals
Scope	Proposal Management
Level	Subfunction
Primary Actor	System (for debugging)
Main Success Scenario	Returns raw file lines

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Name: Rana Asad Ur Rahman

Registration Number: SP23-BSE-029

Fully Dressed Use Case: Update Project Status

Use Case Name	Update Project Status
Scope	Project Management System
Level	User goal
Primary Actor	Faculty
Stakeholders and Interests	- **Faculty**: Wants to reflect the accurate status of the project based on its progress. - **Students**: Want to be informed of their project's progress. - **Admin**: Needs accurate records for tracking and reporting.
Preconditions	- Project exists in the system. - Faculty is assigned as the advisor to the project. - User is authenticated.
Post conditions	- Project status is updated successfully in the system.
Success Metrics	The status of the project is updated to the selected new status and stored correctly.
Minimal Guarantee	The system logs the update attempt even if it fails and shows an appropriate error message.

📋 Main Success Scenario (Basic Flow)

1. Faculty logs into the system.
2. Faculty navigates to the project list and selects a specific project.
3. Faculty clicks on the “Update Status” button.
4. The system displays current project status and a list of valid status options (e.g., "Not Started", "In Progress", "Completed").
5. Faculty selects a new status and clicks “Submit”.
6. The system validates the change.
7. The system updates the project status in the database.

8. The system confirms the update with a success message.
9. Project status is now reflected in the project details.

🔁 Extensions (Alternate Flows)

3a. Project not found

- 3a1. System shows an error: "Project not found."
- 3a2. Use case ends.

5a. Invalid status selected

- 5a1. System shows an error: "Invalid status."
- 5a2. Faculty selects a valid status.
- 5a3. Returns to step 6.

6a. Database update fails

- 6a1. System displays error: "Failed to update project status."
- 6a2. Faculty retries or contacts admin.
- 6a3. Use case ends.

🔐 Special Requirements

- The system must validate user permissions before allowing the update.
- Status values must be predefined and consistent (possibly from an enum).
- Audit trail should log the date/time of the update and the actor's ID.

Zarak Khan

Comprehensive Use Case Specification for Admin in FYP Management System

1. System Overview

The **Final Year Project (FYP) Management System** allows university administrators to manage student projects, supervisors, evaluations, and system configurations. The **Admin** is the primary actor responsible for maintaining data, generating reports, and ensuring smooth system operations.

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2. Actors

Actor	Description
Admin	Manages students, supervisors, projects, and system settings.
Student	(Secondary) Submits project proposals and reports.
Supervisor	(Secondary) Guides students and evaluates projects.

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3. Detailed Use Cases

Use Case 1: Login to System

- **Actor:** Admin
- **Description:** Admin logs into the FYP Management System.
- **Preconditions:** Admin has valid credentials.
- **Basic Flow:**

1. Admin enters email and password.
2. System verifies credentials.
3. On success, redirects to Admin Dashboard.

- **Alternative Flow:**
  - If credentials are invalid, system displays an error.

Post conditions: Admin gains access to the system.

Use Case 2: Manage Students

- **Actor:** Admin
- **Description:** Admin adds, edits, or removes student records.
- **Preconditions:** Admin is logged in.
- **Basic Flow:**

1. Admin clicks **"Manage Students"**.
2. System displays a list of students.
3. Admin selects:

- **Add Student**: Enters (Name, ID, Email, and Program).
- **Edit Student**: Modifies existing details.

- **Delete Student**: Removes record after confirmation.

4. System validates and updates the database.

- **Alternative Flows:**
  - If student ID already exists, system rejects duplication.

Post conditions: Student records are updated.

Use Case 3: Manage Supervisors

- **Actor:** Admin
- **Description:** Admin adds or assigns faculty supervisors.
- **Preconditions:** Admin is logged in.
- **Basic Flow:**

1. Admin clicks **"Manage Supervisors"**.
2. System shows the list of supervisors.
3. Admin:
   - **Adds Supervisor** (Name, Department, Max Projects).
   - **Assigns Supervisor** to a project.
4. System checks availability and updates records.

- **Alternative Flows:**
  - If supervisor's project limit is exceeded, system shows an error.

Post conditions: Supervisor assignments are updated.

Use Case 4: Manage Projects

- **Actor:** Admin
- **Description:** Admin creates, assigns, and tracks FYP projects.
- **Preconditions:** Students and supervisors are registered.
- **Basic Flow:**

1. Admin clicks **"Manage Projects"**.
2. Chooses:
   - **Create Project** (Title, Description, and Deadline).

- **Assign Student & Supervisor** (from dropdown lists).

3. System validates and links them in the database.

- **Alternative Flows:**
  - If a student is already assigned, system prevents duplication.

Post conditions: Project is added and visible in tracking.

Use Case 5: Generate Reports

- **Actor:** Admin
- **Description:** Admin exports project progress or evaluation reports.
- **Preconditions:** Projects exist with submission data.
- **Basic Flow:**

1. Admin clicks **"Generate Reports"**.
2. Selects report type:
   - **Progress Report** (Student milestones).
   - **Evaluation Report** (Supervisor feedback).
3. Filters by Department/Date Range.
4. System compiles data into **PDF/Excel**.
5. Admin downloads or prints.

- **Alternative Flows:**
  - If no data exists, system displays "No records found."

Post conditions: Report is generated for review.

Use Case 6: System Configuration

- **Actor:** Admin
- **Description:** Admin sets deadlines, grading rules, and permissions.
- **Basic Flow:**

1. Admin clicks **"System Configuration"**.
2. Updates:
   - **Submission Deadlines** (e.g., Proposal: 30-May-2024).

- **Grading Criteria** (e.g., 70% for final report).

3. System saves changes and notifies affected users.

- **Alternative Flows:**
  - If a deadline is in the past, system rejects it.

Postconditions: System settings are updated.

Name: Abdullah

Reg No: SP23-BSE-116

Use Case 1: addMember()

Use Case Name: Add Member

Scope: Group Management

Level: User goal

Primary Actor: Administrator

Stakeholders and Interests:

Administrator: Wants to add students to a group efficiently.

Students: Want to be part of the correct group for their semester project.

Preconditions:

- The group must already exist.
- The student must be registered in the system.
- The student is not already part of another group.

Postconditions:

The student is successfully added to the group's member list.

Main Success Scenario (Basic Flow):

- Administrator selects an existing group.
- Administrator selects a student to add.
- System checks if the student is already part of any group.
- System adds the student to the selected group.
- System confirms the addition and displays updated group information.

Extensions (Alternate Flows):

3a. If the student is already part of another group:

→ System shows an error message and aborts the addition.

4a. If the group has reached its member limit:

→ System prevents the addition and notifies the administrator.

Special Requirements:

- Real-time validation of student eligibility.
- Notification sent to the student upon successful addition.

✅ **Use Case 2:** removeMember()

Use Case Name: Remove Member

Scope: Group Management

Level: User goal

Primary Actor: Administrator

Stakeholders and Interests:

Administrator: Needs control over group membership.

Students: Should be properly removed if they withdraw or switch groups.

Preconditions:

- Group and student must exist.
- Student must already be a member of the group.

Postconditions:

The student is removed from the group member list.

Main Success Scenario (Basic Flow):

- Administrator opens group details.
- Administrator selects the student to remove.
- System verifies the student's membership in the group.
- Student is removed from the list.
- System confirms removal.

Extensions (Alternate Flows):

3a. If the student is not found in the group:

→ System notifies the admin and aborts the removal.

4a. If the removed student is a group leader:

→ System prompts to assign a new leader or continue.

Special Requirements:

Notification sent to the student upon removal.

✅ **Use Case 3:** assignAdvisor()

Use Case Name: Assign Advisor

Scope: Group Management

Level: User goal

Primary Actor: Administrator

Stakeholders and Interests:

Administrator: Needs to assign supervisors fairly and efficiently.

Supervisors: Want to be assigned groups within their capacity.

Students: Need an advisor for guidance.

Preconditions:

- Group must exist.
- Advisor must be registered.
- Advisor must be available (not exceeding their group limit).

Postconditions:

Advisor is successfully linked to the group.

Main Success Scenario (Basic Flow):

- Administrator selects a group.
- Administrator chooses an advisor from a list.
- System checks advisor's availability.
- Advisor is assigned to the group.
- Confirmation message is shown.

Extensions (Alternate Flows):

3a. If the advisor already has maximum assigned groups:

→ System blocks the assignment and shows a warning.

Special Requirements:

- System should prevent duplicate advisor assignments to the same group.
- Email notification to advisor and group members upon assignment.

fully Dressed Use Case: Submit Proposal

1. Basic Information*

Element *Description* |

Use Case Name | Submit Proposal |

Actor | Student |

2. Preconditions

- Student is logged in.
- Student has a registered group.
- Submission period is open.

3. Main Success Scenario

1. Student selects "Submit Proposal".
2. System displays a submission form (title, description, file upload).
3. Student fills details and uploads a proposal file (PDF/DOCX).
4. System validates the file (format, size ≤10MB).
5. Supervisor is notified; proposal status changes to "Under Review".

4. Alternative Flows

- *A1: Invalid File
  - Ste 4a: System rejects non-PDF/DOCX files.

Aizaz Ullah ->Fully Dressed

Use Case Element	Details
Use Case Name	Login
Scope	Authentication System
Level	User Goal
Primary Actors	Student, Group Leader, Supervisor
Stakeholders and Interests	- Student/Group Leader/Supervisor: Want to access the system securely.
	- System: Ensures only valid users are authenticated.
Preconditions	User must already be registered in users.txt.
Postconditions (Success)	User is authenticated and redirected to their dashboard.
Postconditions (Failure)	User remains on login screen with an error message.
Main Success Scenario	1. User opens the login interface.
	2. User enters role, ID, and password.
	3. System reads users.txt.
	4. If a match is found, user is authenticated.
	5. System redirects user to respective dashboard.
Extensions (Alternate Flows)	3a. File not found – system logs error and displays error message.
	4a. Invalid credentials – user is shown an error and stays on login screen.
Special Requirements	- Password matching must be case-sensitive.
	- File I/O must be reliable (users.txt must exist and be accessible).

Fully Dressed use case

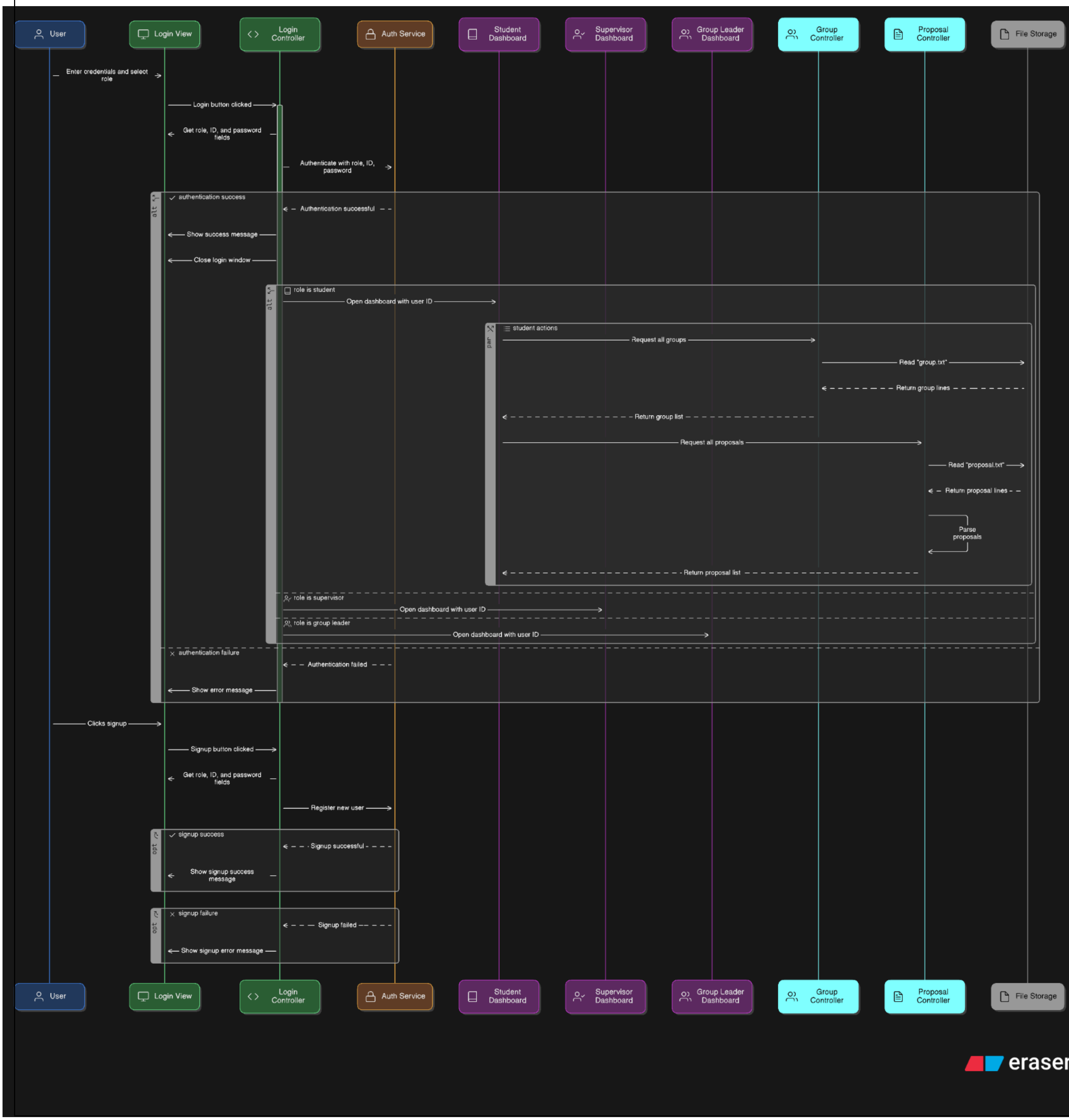
Ahamd Ayyar Khan

Use Case Element	Details
Use Case ID	UC-SUP-03
Use Case Name	Evaluate Submissions
Primary Actor	Supervisor
Stakeholders and Interests	Supervisor (wants to grade fairly and efficiently), Students (want timely feedback)
Preconditions	- Supervisor must be logged in. - The project must be assigned to the supervisor.
Postconditions	- Submission is marked as evaluated. - Grades and feedback are saved in the system.
Main Success Scenario	1. Supervisor logs into the system. 2. Navigates to assigned projects. 3. Selects a submission. 4. Views content of the submission. 5. Enters feedback and grade. 6. Clicks "Submit Evaluation". 7. System stores evaluation and notifies student.
Extensions (Alternative Flows)	3a. No submissions available: System shows message "No pending submissions". 5a. Supervisor cancels evaluation: System returns to project list without saving.
Special Requirements	- Evaluation should support file previews (PDF, DOCX). - Should allow file attachments in feedback.
Frequency of Use	Weekly or bi-weekly during submission periods.
Business Rules	- Grades must be within a valid range (e.g., 0–100). - Feedback is mandatory for final evaluation.

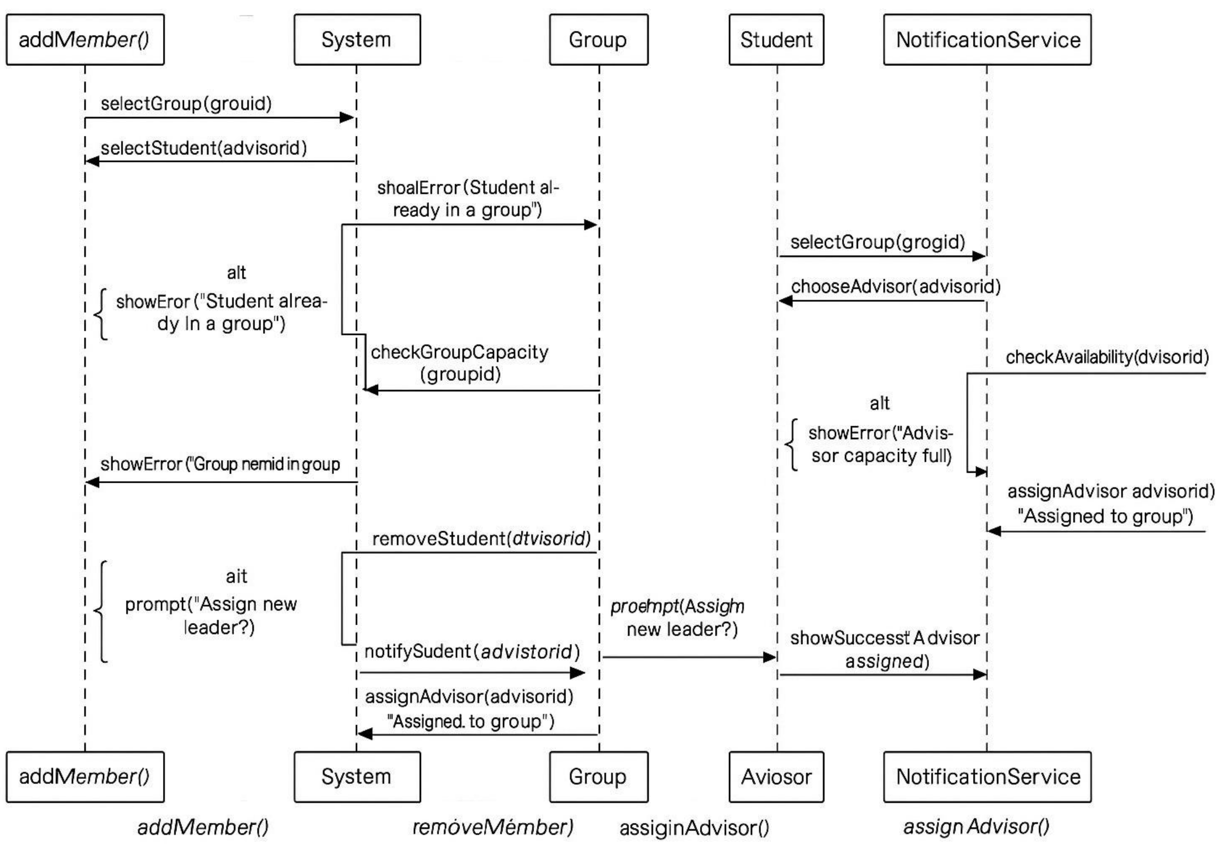
Chapter 3

Sequence Diagrams

Mustafa



Abdullah


addMember()
System
Group
Student
NotificationService
selectGroup(grouid)
selectStudent(advisorid)
shoalError(Student al-
ready in a group")
selectGroup(grogid)
alt
showEror ("Student alrea-
dy In a group")
chooseAdvisor(advisorid)
checkGroupCapacity
(groupid)
checkAvailability(dvisorid)
alt
showError("Advis-
sor capacity full)
showError ("Group nemid in group
assignAdvisor advisorid)
"Assigned to group")
removeStudent(dtvisorid)
ait
prompt("Assign new
leader?)
proempt(Assigm
new leader?)
showSuccesst Advisor
assigned)
notifySudent(advistorid)
assignAdvisor(advisorid)
"Assigned. to group")
addMember()
System
Group
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NotificationService
addMember()
removeMember)
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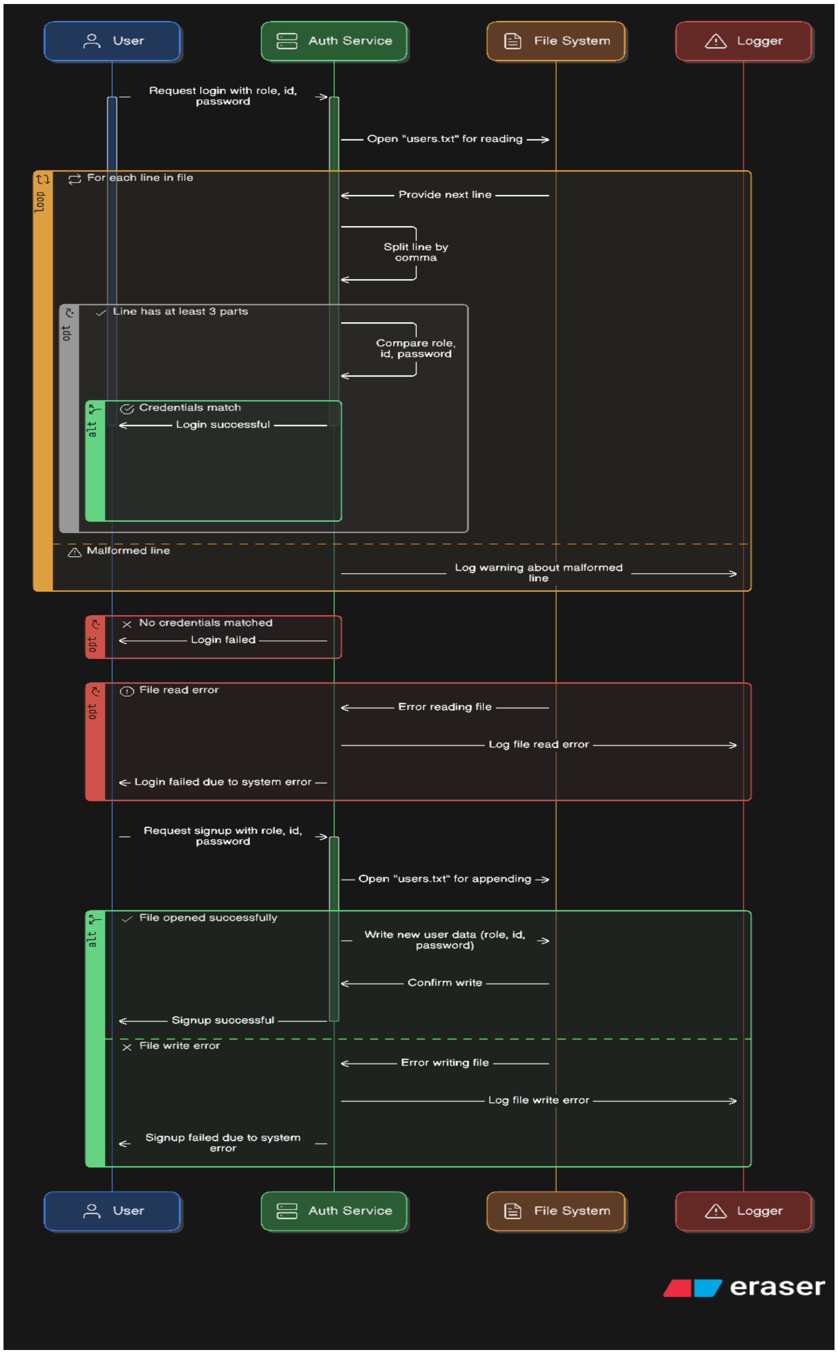
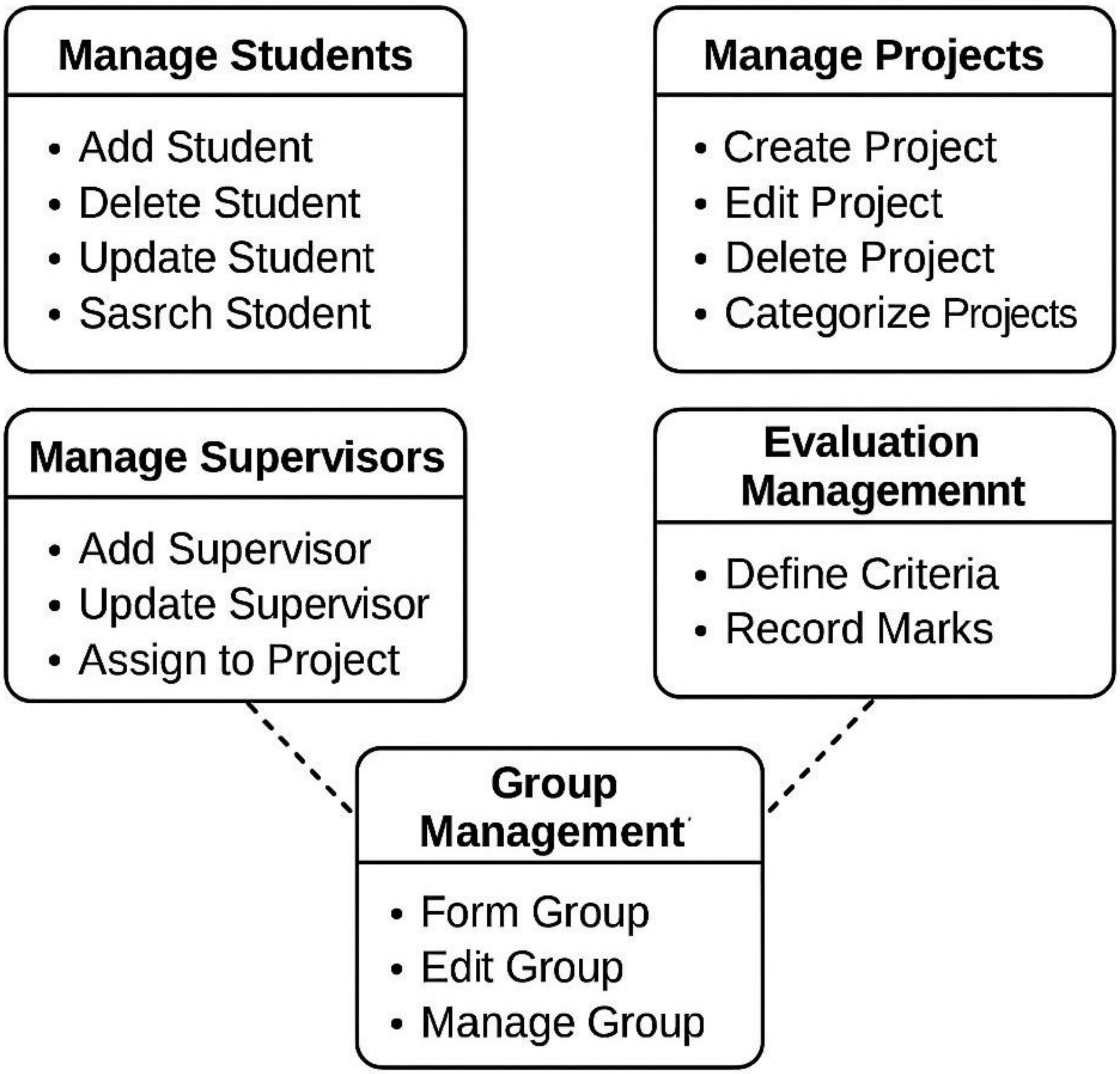
assign Advisor()

Aizaz Ullah


User
Auth Service
File System
Logger
Request login with role, id, password
Open "users.txt" for reading
loop
For each line in file
Provide next line
Split line by comma
opt
Line has at least 3 parts
Compare role, id, password
alt
Credentials match
Login successful
Malformed line
Log warning about malformed line
opt
No credentials matched
Login failed
opt
File read error
Error reading file
Log file read error
Login failed due to system error
Request signup with role, id, password
Open "users.txt" for appending
alt
File opened successfully
Write new user data (role, id, password)
Confirm write
Signup successful
File write error
Error writing file
Log file write error
Signup failed due to system error
User
Auth Service
File System
Logger
eraser

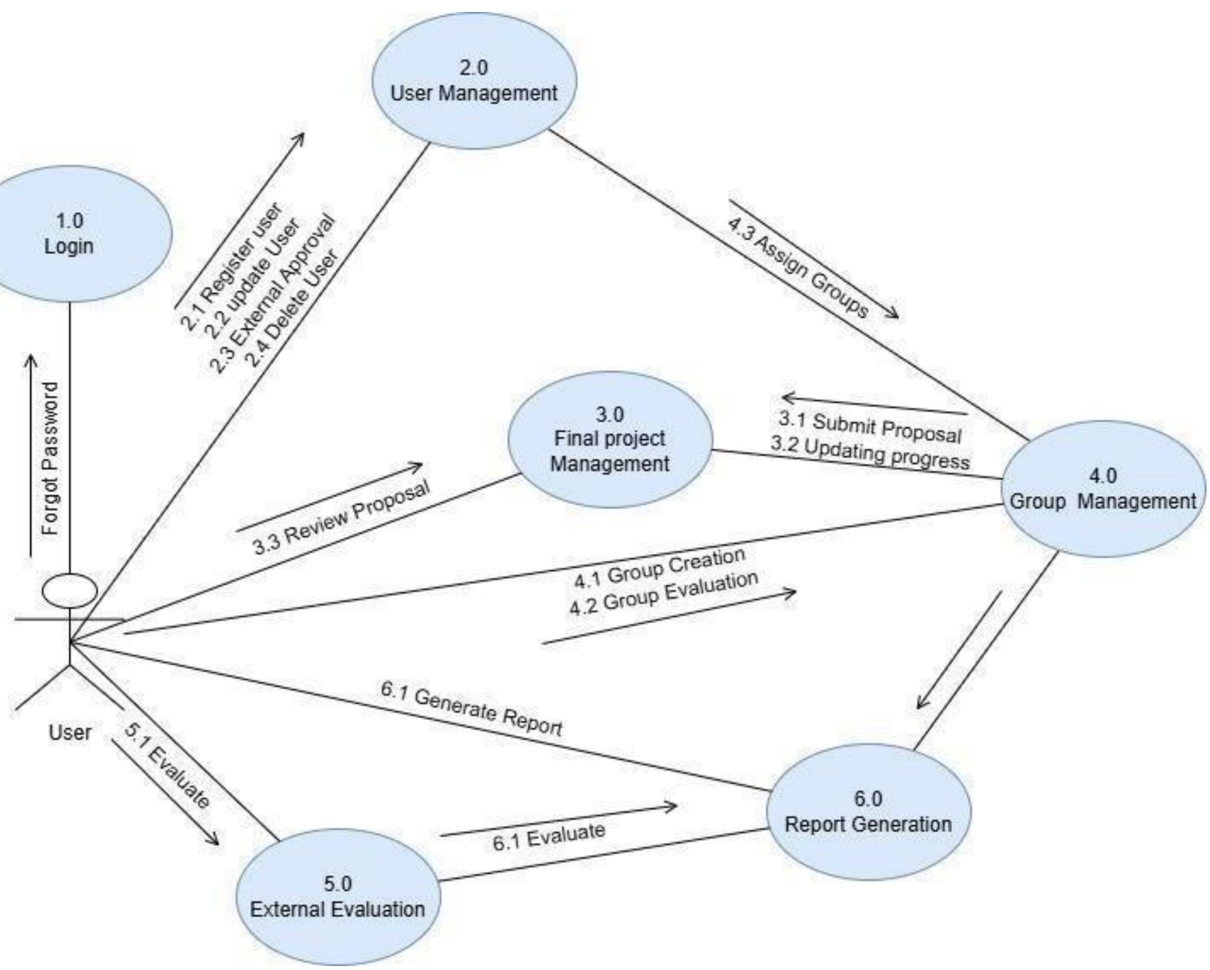
Chapter 4



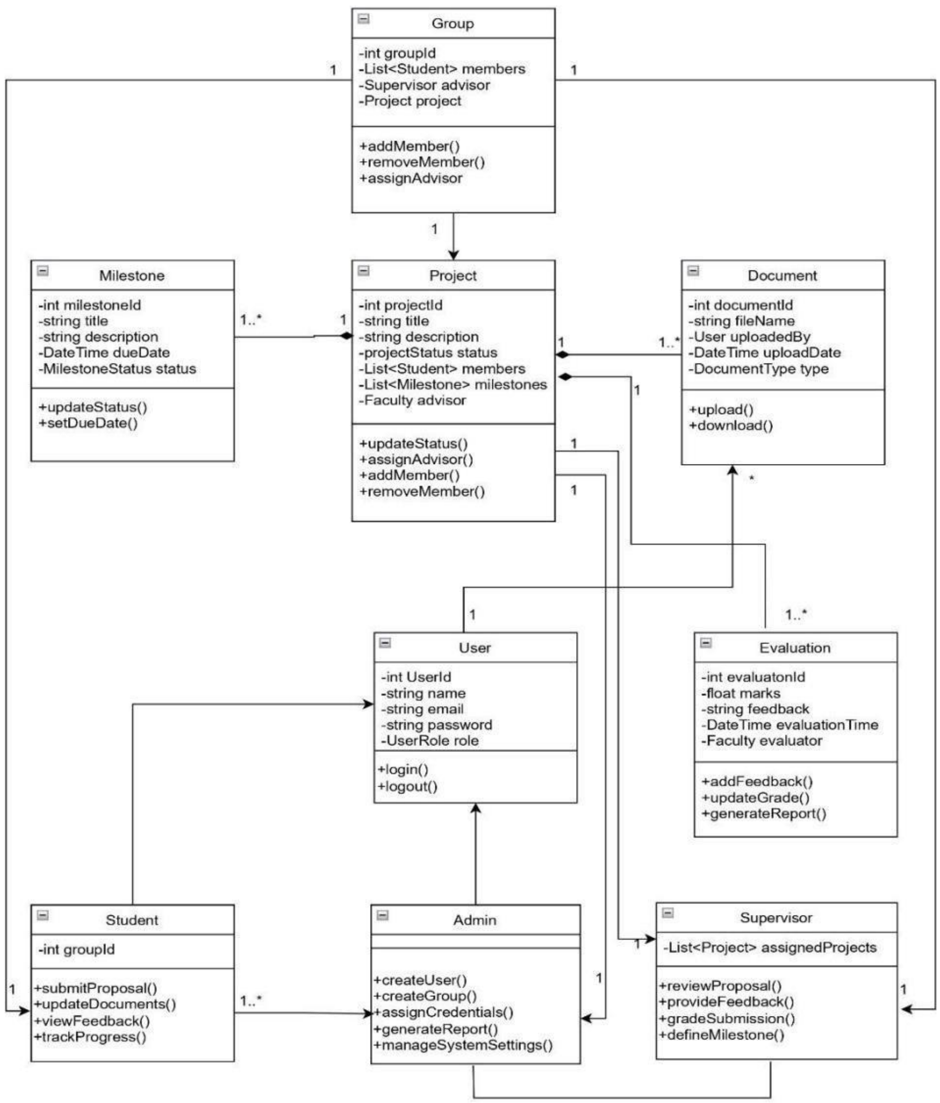
Package Diagram anrurm

Chapter 5

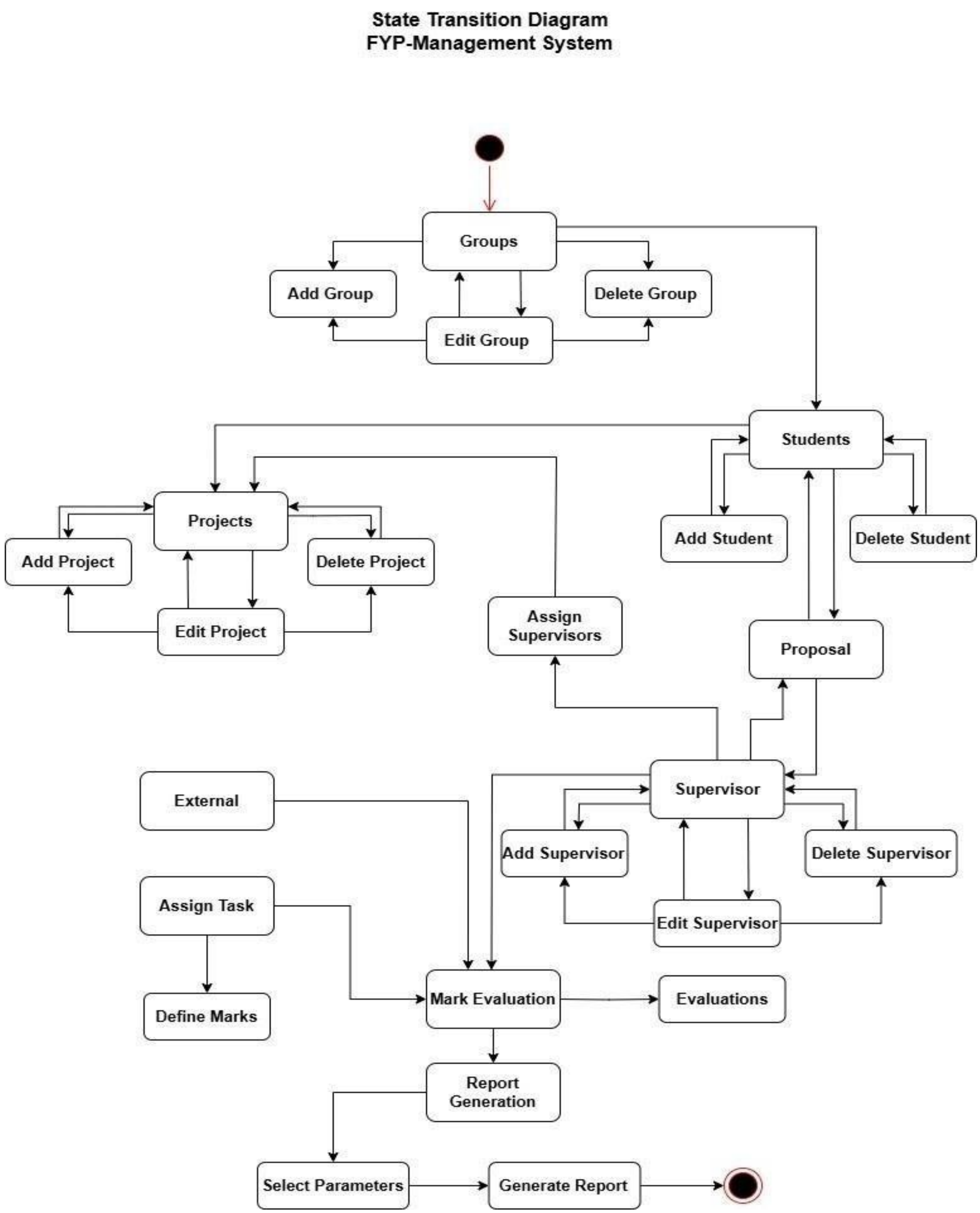
Collaboration Diagram



Chapter 6


Group
-int groupId
-List<Student> members
-Supervisor advisor
-Project project
+addMember()
+removeMember()
+assignAdvisor
Milestone
-int milestoneId
-string title
-string description
-DateTime dueDate
-MilestoneStatus status
+updateStatus()
+setDueDate()
Project
-int projectId
-string title
-string description
-projectStatus status
-List<Student> members
-List<Milestone> milestones
-Faculty advisor
+updateStatus()
+assignAdvisor()
+addMember()
+removeMember()
Document
-int documentId
-string fileName
-User uploadedBy
-DateTime uploadDate
-DocumentType type
+upload()
+download()
User
-int UserId
-string name
-string email
-string password
-UserRole role
+login()
+logout()
Evaluation
-int evaluatonId
-float marks
-string feedback
-DateTime evaluationTime
-Faculty evaluator
+addFeedback()
+updateGrade()
+generateReport()
Student
-int groupId
+submitProposal()
+updateDocuments()
+viewFeedback()
+trackProgress()
Admin
+createUser()
+createGroup()
+assignCredentials()
+generateReport()
+manageSystemSettings()
Supervisor
-List<Project> assignedProjects
+reviewProposal()
+provideFeedback()
+gradeSubmission()
+defineMilestone()
1
1..*
*

State Transition Diagram



Chapter 7

Coding Standards for FYP Group Management System

1. Class Naming Convention

- All class names must use **PascalCase** format.
- Classes must be named based on their responsibilities:
    - Student, Advisor, Group, Administrator, System, NotificationService.

2. Method Naming Convention

- All method names must use **camelCase**.
- Method names must clearly reflect the operation they perform:
    - addMember, removeMember, addStudent, removeStudent, checkStudentEligibility, assignAdvisor, sendNotification.

3. Access Modifiers and Encapsulation

- All class fields must be declared as private.
- Access to fields must be provided through public getters and setters.
- Only relevant classes may access internal methods marked as private.

4. Constructor Standards

- Each class must define a constructor to initialize required fields.
- No business logic is allowed inside constructors.

5. Method Structure

- Each method must perform a single, clearly defined task.
- Each method must include:
    - Input validation.
    - Null checks before object operations.
    - Clear return values or appropriate exception handling.

6. Collection Handling

- Collections such as List<Student> must be initialized using concrete classes like ArrayList.
- Iteration through collections must use enhanced for loop or iterator.

7. Code Formatting Rules

- Indentation must be 4 spaces per level.
- Curly braces {} must be placed on the same line as the control structure or method name.
- A single blank line must be used between methods.

8. Notification Service Usage

- NotificationService.sendNotification(String message, Student recipient) must be called:
  - When a student is added to a group.
  - When a student is removed from a group.
  - When an advisor is assigned to a group.

9. Class Responsibilities

- Administrator class is only responsible for managing group membership (adding/removing students).
- Group class maintains a list of Student members and an optional Advisor.
- System class is responsible for:
  - Checking student eligibility.
  - Assigning advisors.
- NotificationService only handles notification logic.

10. Exception Handling

- All invalid operations must throw standard Java exceptions such as IllegalArgumentException.
- No silent failures are allowed.

11. Naming Standards

- Variables must be named clearly and meaningfully.
  - Examples: student, group, advisor, message, recipient.
- No abbreviation of names unless they are universally accepted (e.g., ID).

12. Comments and Documentation

- All public classes and methods must include JavaDoc comments.
- Inline comments must be used sparingly and only where logic is not self-explanatory.